Generation Technologies

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Contents

1 Introduction

This note is a catalogue. It collects, in one place, what the rest of the course assumes you know about the machines that produce electricity and heat: what they are, what they cost, and what their costs imply about how they are used. It solves no model and states no optimisation problem. When a lecture says “a peaking plant” or “a capital-intensive technology” and moves on, this is the note it is moving on from.

A word of reassurance before we start, because the subject matter looks technical and mostly is not. This is an economics course, and I am an economist. We will not explain how a turbine works; what we need from a technology is much narrower than what an engineer needs from it, and Section 2 is devoted to saying how narrow.

1.1 What a generation technology is

Let us fix some vocabulary. **Primary energy** is energy in the form we find it: coal in a seam, gas in a reservoir, uranium in ore, wind in the atmosphere, sunlight, water held behind a dam. **Secondary energy** is energy in a form we have made in order to use it: electricity above all, and also district heat and hydrogen. An **energy service** is what we actually want — a warm room, a cold beer, a moving car — and getting from secondary energy to a service takes yet another piece of equipment, the radiator, the fridge, the motor.

A **generation technology** is the capital that converts primary energy into secondary energy. That is the whole definition, and everything in Section 3 is an instance of it. Two remarks are worth making up front. First, we’ll focus mostly on conversion: Primary energy is a commodity and mostly behaves like one; energy services are consumption and mostly behave like that. The middle step that requires a large, long-lived, immovable, single-purpose piece of capital makes energy markets peculiar.

Second, a **battery** converts nothing. It takes electricity in and gives electricity back later, minus a little. It appears in this note anyway, in Section 3.9, because it competes with generation for the same job at the same times.

1.2 Units

Power is not energy. A capacity is a rate, measured in watts and its multiples (kW, MW, GW). An energy is a rate multiplied by a duration, measured in watt-hours (kWh, MWh, GWh). A 1 MW turbine running flat out for a year produces 8,760 MWh. An investment cost is quoted per unit of *capacity* (EUR/kW); a fuel cost or a levelised cost is quoted per unit of *energy* (EUR/MWh).

Thermal energy is not electrical energy. A plant that burns fuel consumes MWh of *thermal* energy and produces MWh of *electrical* energy, and there are always more of the former. Where it matters I write MWh_{th} and MWh_{e}. The exchange rate between them is the plant’s efficiency.

Money has a year. When we reference euros in the note, we use the base year 2020.[1] Appendix A carries the conversion factors and a worked example.

1.3 Where these numbers are from, and when

Throughout the note, **most numbers are Danish**, or near enough: Roughly three fifths of the parameters in the cost tables come, through the compilation described in Appendix D, from the Danish Energy Agency's technology catalogue — a free, comprehensive, machine-readable and regularly updated source, and for those reasons the one most of the European modelling world uses. On top of that, the reference capacity factors are roughly what Danish sites deliver, and the fuel prices are European. So when this note says that a wind turbine costs a certain amount, it means in Denmark, at Danish permitting, on a Danish site, financed on European terms.

In general, unless otherwise stated, estimates are from 2025. Technology costs and estimates, however, vary a lot over time and space. Sections Section 5 and Section 6 deal with this explicitly.

1.4 How the note is built

Section 2 sets up the one framework the note uses: four kinds of cost, and the handful of technical parameters that generate them. Section 3 is the tour. Each technology gets the same card with the same three slots (what it is, what its costs look like, what limits its deployment), so that you can compare technologies by reading down a column rather than re-reading prose. The tour ends with two summary tables and a figure with estimates. Section 4 takes the single number everyone quotes — the levelised cost of energy — and asks what it does and does not say. Section 5 and Section 6 are the two short sections promised above: what happens to costs over time, and why the same machine costs different amounts in different places. Both are applications of Section 4. Section 7 closes by mapping cost structure onto operational role: dispatchable and variable, baseload and peaker, storage, co-generation.

Appendices keep definitions, elaborate on a few things from the note, keep records of where data is collected from and how to reproduce them. Specifically, appendix C collects terms the note defines and include a list of terms it does not, but which you will have met elsewhere in the course.

Every number quoted in the text is computed by the scripts in the Github repository from data fetched at a pinned version, so the text, the tables and the figures cannot disagree with each other. And every number is either sourced or explicitly labelled as an assumption of mine — Appendix D keeps the score.

[1] The dataset is PyPSA's `technology-data` at tag v0.13.2 (PyPSA, Danish Energy Agency, et al. 2025), which compiles the Danish Energy Agency catalogue (Danish Energy Agency and Energinet 2024) together with several other published sources and converts everything to a common currency year. Appendix D says which numbers come from where.

2 Plant characteristics and the decomposition of costs

One plant, one cost, four ways of paying it. This section sets up the taxonomy that the rest of the note applies: Section 3 fills it in machine by machine, Section 4 collapses it into a single number, and Section 7 unpacks it again into an operational role.

2.1 Four kinds of cost

Power plants are conventionally described by four cost categories. They differ in what they are proportional to and in when they are paid, and those two differences are what does all the work later.

Investment costs. What it costs to build the plant, quoted in EUR per kW of capacity. Paid up front, or spread over the construction period, and roughly proportional to the size of the plant. The important feature is not that it is large but that it is largely *sunk*.

Fixed operating and maintenance costs. What it costs to keep the plant available for a year — staff, insurance, scheduled maintenance — quoted either in EUR per kW per year or, as in the data used here, as a percentage of the investment cost per year. Also proportional to capacity, and paid whether or not the plant produces a single MWh.

Variable costs. What it costs to produce one more MWh: fuel, plus a small variable operating cost for wear and consumables. Quoted in EUR per MWh and proportional to output. This is the cost that becomes the plant's marginal cost in any dispatch model, and hence the one that determines the order in which plants are used — the **merit order**, in which the cheapest available plant runs first and the most expensive one needed runs last.

Adjustment costs. What it costs to *change* output — starting up from cold, ramping quickly, running below the minimum load at which the plant is stable, or cycling on and off day after day. Unlike the other three, this is not proportional to anything convenient. It is a cost of a *pattern* of operation rather than of a level, which makes it tricky to summarize and compare.

Fuel, efficiency and carbon

For a plant that relies on fuel injection, the variable cost structure has a simple structure. Let p^F be the price of fuel per MWh of *thermal* energy, let η be the plant's conversion efficiency — the share of the fuel's energy that comes out as electricity — and let o be the variable operating cost. Then the marginal cost of a MWh of electricity is

$$c = \frac{p^F}{\eta} + o. \tag{1}$$

If the fuel carries φ tonnes of CO_2 per MWh of thermal energy, the same MWh of electricity carries

$$e = \frac{\varphi}{\eta} \tag{2}$$

tonnes. Notice the double role of η: it sits in the denominator of both expressions, so an efficient plant is cheap *and* clean per MWh delivered.

A worked example, since the formula is easier to believe with numbers in it, and we may as well use the note's own coal plant so that the answer is one you will meet again. Its efficiency is $\eta = 0.356$, it pays 12 EUR per $\mathrm{MWh_{th}}$ for coal, and its non-fuel operating cost is 3.26 EUR per MWh. Then

$$c = \frac{12}{0.356} + 3.26 = 37.0 \text{ EUR/MWh}, \qquad e = \frac{0.336}{0.356} = 0.94 \text{ tCO}_2\text{/MWh},$$

which is the coal row of Table 2. You may verify that halving the efficiency to 0.178 raises the marginal cost to 70.7 and doubles the emission rate to 1.89: the same fuel, at the same price, in a different machine.

2.2 The technical parameters underneath

Behind the four cost categories sits a short list of engineering quantities. None of them requires any physics to use; each is a number you look up.

- **Conversion efficiency** η, the ratio of energy out to energy in. Between about 0.2 and 0.6 for plants that burn things, and undefined for wind and solar, which have no fuel to convert.
- **Carbon content of the fuel** φ, in tonnes of CO_2 per $\mathrm{MWh_{th}}$. A property of the fuel, not of the plant.
- **Nameplate capacity**, the output the plant is rated for. Worth distinguishing from *available* capacity, which is what is actually there after outages and weather.
- **Availability** $\gamma \in [0, 1]$, the share of nameplate capacity available in a given hour. For a thermal plant this is close to one except when it is broken or under maintenance; for a wind farm it is whatever the weather is doing.
- **Technical lifetime**, from 10 years for a battery inverter to 80 for a hydro scheme. Together with the discount rate it decides how heavily the investment cost weighs on each MWh.
- **Ramp rate, minimum load and start-up time**, the parameters behind adjustment costs.

Capacity factor

The **capacity factor** is the energy a plant actually produced over some period, divided by what it would have produced running flat out for the whole period:

$$CF = \frac{\sum_h E_h}{\bar{q} \cdot H}, \tag{3}$$

where $\bar{q}$ is capacity and H the number of hours. Equivalently, and often more intuitively, a technology's **full-load hours** are $CF \times 8{,}760$: the number of hours a year it would have run had it run flat out or not at all.

For a dispatchable plant the capacity factor is an *outcome of an economic decision*: the plant runs when it is worth running. For a wind farm or a solar park it is largely a *statement about the site*: the resource arrives, or it does not.

2.3 General characteristics of power plants

Four features are shared widely enough across the fleet to be worth stating as generalities, each with the economic consequence attached.

There is very little fuel substitution after the fact. A coal-fired boiler cannot burn wood chips without substantial new capital, and a gas turbine cannot burn coal at all. The fuel is chosen when the plant is built, not when it runs. This is why models treat fuel as an attribute of a technology rather than as a choice, and why a fuel price shock changes which plants run rather than what they burn.

Time-to-build is long, and very unevenly so. Solar parks take months. Wind farms and gas plants take a few years, most of it permitting. Nuclear takes the better part of a decade at best, and considerably longer in Europe and North America than in China. The consequence is not subtle: a capacity path that requires a lot of the slow technologies cannot be walked quickly, whatever it costs.

These are capital-intensive assets. Most of the lifetime cost is committed before a single MWh has been sold, into an asset that cannot be moved and has no alternative use. That combination is what makes the cost of capital a first-order determinant of the cost of energy, which is the subject of Section 4.3, and it is why so much energy policy is in substance about reducing revenue risk.

Plants are lumpy. They come in discrete sizes — you cannot buy 0.3 of a reactor — and the models we use elsewhere in the course pretend otherwise. The approximation is usually harmless at the scale of a national system and badly wrong at the scale of a single investment decision.

2.4 So what do we actually need from a technology?

To put a technology into an economic analysis of the energy system you need four things, and mostly not much else:

i. what it costs to build and keep (investment and fixed O&M, per kW);
ii. what it costs to run (fuel and variable O&M, per MWh, which Equation (1) builds out of a fuel price and an efficiency);
iii. how freely it can change output, and at what cost;

iv. how much of its capacity is available, when.

The whole of Section 3 is an attempt to fill in those four for one machine at a time.

3 A tour of the technologies

Let us now put machines into the framework. Every technology below gets the same card, with the same three slots, in the same order:

- **What it is** — what goes in, what comes out, and the one or two features that matter economically.
- **Cost signature** — which of the four categories of Section 2.1 dominate, and why.
- **What limits deployment** — fuel, siting, land, grid, materials, acceptance.

The first slot is deliberately thin. For how the machines actually work, the technology catalogues cited in Appendix D are the place to look. All the numbers quoted below are collected in Table 2 at the end of the section.

3.1 Thermal plants

Five of the technologies below — coal, gas, biomass, waste and nuclear — are thermal plants: heat is released from a fuel, the heat drives a turbine, and the turbine drives a generator. In a nuclear plant the heat comes from fission rather than combustion, and nothing else about the arrangement changes. Three consequences follow.

i. **Most of the fuel's energy does not become electricity.** A typical plant converts between a fifth and three fifths of it; the remainder leaves as waste heat, usually up a chimney or into a river. That share is the efficiency η of Equation (1), and the waste is what co-generation (Section 3.10) recovers.

ii. **The machine is hot and heavy, so it cannot be started, stopped or turned down instantly.** Heating several hundred tonnes of steel takes hours and costs fuel, and doing it repeatedly wears the plant out faster. This is the origin of the adjustment costs of Section 2.1.

iii. **The fuel is bought continuously**, so these plants have a real marginal cost. The technologies from Section 3.6 onwards do not.

3.2 Gas turbines

Gas turbines: the peaker and the combined cycle

What it is Burn gas, spin a turbine. The simple version, an **open-cycle** gas turbine (OCGT), lets the hot exhaust go and converts 40% of the fuel. The elaborate version, a **combined-cycle** plant (CCGT), adds a second stage that uses the exhaust heat to raise steam, and reaches 57%. Both start quickly by the standards of thermal plant; the simple one starts in minutes.

Cost signature The extra machinery of the combined cycle costs money to build and saves fuel, so the peaker is cheap to build and expensive to run, and the combined cycle is the reverse, from the same fuel at the same price. In our numbers the CCGT costs 905 EUR/kW against the peaker's 470, roughly twice as much, and has a marginal cost of 66 EUR/MWh against the peaker's 91. That is a choice made at the investment stage, not a fact about gas.

What limits deployment Gas supply and its price, which has proved more volatile than anyone assumed before 2021; carbon pricing; and the question of building a twenty-five-year fossil asset into a system that intends to decarbonise within its lifetime.

3.3 Coal

Coal: the twentieth-century baseload plant

What it is Pulverise coal, burn it, raise steam, spin a turbine. Efficiency 36%. Slow to start, many hours from cold, and slow to change output.

Cost signature Expensive to build, moderate to run, and by a wide margin the dirtiest per MWh in the catalogue: about 37 EUR/MWh of marginal cost, and 0.94 tonnes of CO_2 for every MWh of electricity. At a carbon price of 80 EUR per tonne that adds 76 EUR/MWh, more than doubling the marginal cost, which is why a carbon price moves coal down the merit order faster than anything else.

What limits deployment In Europe, policy. The cost structure that made coal the baseload technology of the twentieth century — high capital, low-to-moderate fuel, poor flexibility — is badly matched to a system that now needs its thermal plants to move around.

3.4 Nuclear

Nuclear: capital, and almost nothing else

What it is The same steam plant with a different heat source. Very expensive to build — our figure is 8,594 EUR/kW against the coal plant's 3,827, the highest of anything here built purely for electricity — very cheap to fuel, very long-lived, and very slow to build.

Cost signature Almost entirely capital. The marginal cost is about 14 EUR/MWh, so there is almost nothing to save by turning the plant down: whatever a reactor is technically capable of, its economics point at running flat out. Because so much of the cost is committed so far ahead of the revenue, nuclear's levelised cost depends on the discount rate more than anything else in the table; Section 4.3 shows how much.

What limits deployment Construction cost and schedule, which have tended to worsen rather than improve in Europe and North America: Grubler (2010) documents the French programme as a case of *negative* learning by doing. Decommissioning and long-term waste management are real costs that sit outside the four categories, because they are paid long after the plant stops earning.

3.5 Biomass and waste

Biomass and waste: costly fuel, favourable accounting

What it is A steam plant again, burning wood chips, wood pellets or sorted municipal waste. Electrical efficiencies are low, 21% for the waste plant in our table and 27% for the biomass one, because these are built to produce heat as well (Section 3.10).

Cost signature Expensive to build and expensive to fuel: judged on electricity alone, biomass is weak on both counts. Its carbon is conventionally counted as zero, which is what makes it attractive under a carbon price.[a] Waste is the oddity: the plant is paid to accept its fuel, so the fuel price is effectively negative, but it carries the largest variable operating cost in the catalogue, and the electricity is close to a by-product of a waste-disposal service.

Waste and carbon Municipal waste is often counted as carbon-neutral alongside biomass, but roughly half of it by energy content is plastics, and that half is fossil carbon. Divided by a very low electrical efficiency, this gives waste one of the higher emission rates in Table 2. This note counts only the fossil fraction; Appendix D records the assumption.

What limits deployment Fuel supply, and in particular how far the fuel has to travel before the accounting stops looking good; competing uses for the same biomass; and, for waste, the fact that the quantity available is set by how much rubbish there is rather than by how much electricity is wanted.

[a] The accounting treats the carbon released as carbon the plant would have absorbed anyway, which is a claim about the counterfactual use of the land and the timescale of regrowth rather than a measurement, and it is contested.

3.6 Wind

Wind: the site is half the technology

What it is Moving air turns a rotor, which turns a generator. The energy available in the wind rises with the cube of wind speed, so a site with 20% more wind is worth far more than 20% more. That is why turbines have grown so tall, and why site quality matters here in a way it does not for a plant that burns something. Above a certain wind speed the turbine holds at its rated output.

Cost signature Essentially all investment and fixed O&M; the variable cost is a euro or two for maintenance. The plant does not choose when to produce, it collects what is offered, and its capacity factor is a property of the site. Offshore is the same machine, more expensive to build and to maintain, with a higher and steadier capacity factor and a grid connection that is a project in its own right. In our numbers onshore wind comes to 39 EUR/MWh and offshore to 45, close enough that the comparison turns on the assumed site.

What limits deployment Siting and local acceptance, which is now the binding constraint onshore in much of Europe; grid connection; and, at scale, the fact that turbines take wind out of each other's way, so the tenth farm in a region performs worse than the first.

3.7 Solar photovoltaics

Solar PV: no fuel, no moving parts, no choice

What it is Panels turn sunlight into electricity directly. There is no fuel, no fire and nothing that moves, and therefore no start-up cost, no minimum load and nothing to ramp: solar can be turned down instantly and for free. What it cannot do is turn up. Output is whatever the sun provides, which in Denmark is 11.5% of nameplate over a year, concentrated in the middle of the day and the middle of the year.

Cost signature Almost entirely investment. It is also the technology whose investment cost has fallen furthest and fastest (Section 5.2). The gap between utility-scale and rooftop installations is large, 46 EUR/MWh against 89 for the same panels, because everything around them — mounting, wiring, inverters, an electrician on a ladder — costs far more per kW when bought a few tens of kW at a time.

What limits deployment Land, at scale; grid connection; and the fact that every solar park in a region produces at the same time as every other one, which the technology cannot solve on its own.

3.8 Hydroelectricity

Hydro comes in three arrangements that behave differently enough to be described separately.

Hydro: three technologies sharing one name

What it is *Run-of-river* takes the flow as it comes and is therefore a variable technology, closer to wind than to anything dispatchable. *Reservoir* hydro holds water back behind a dam and releases it when it chooses, and is therefore both dispatchable and, in effect, a large seasonal store. *Pumped storage* adds a pump and spends electricity to put the water back up the hill; it generates nothing net and belongs with the batteries.

Cost signature Expensive and highly site-specific to build, almost free to run, extremely long-lived (80 years in our table, against 25 for a gas turbine) and unusually flexible: a hydro plant can go from nothing to full output in minutes, repeatedly, at negligible cost. A reservoir's cost of generating now is not a fuel cost but the value of the water it will not have later. That is an *opportunity* cost, and it is why hydro does not sit still in a merit order the way a gas plant does. The same applies to batteries below.

What limits deployment Geography. There are only so many valleys, most of the good ones in Europe are already used, and public opposition to flooding the remainder is considerable. For the same reason it is a technology with very little scope to get cheaper by being built more often.

3.9 Batteries

Batteries: not a generator

What it is A battery is not a generation technology: it buys electricity and sells it back later, and something like a tenth of it does not come back, which is the **round-trip efficiency**. Two features distinguish it from everything above. First, power and energy are bought separately: a battery is described by a MW rating (how fast it can charge and discharge, essentially the cost of the inverter) and a MWh rating (how much it holds, essentially the cost of the cells), and the ratio between them is its **duration**. Second, the cells degrade with use, so a battery has a cycle life as well as a calendar life.

Cost signature Almost entirely investment, split across the two ratings, and with a shorter life than anything else in the catalogue. As with a reservoir, the cost of discharging now is an opportunity cost rather than a fuel cost.

What limits deployment Materials and supply chains, and above all duration. Four hours of storage is a tool for the daily cycle. It is not a tool for a still, cold fortnight in January, and no reduction in the cost of a four-hour battery makes it one.

3.10 Heat: boilers, heat pumps and co-generation

The course treats heat and electricity as coupled markets, so three heat technologies belong in the tour. They get the same three slots; the only difference is that the product is a MWh of heat, which cannot be compared with a MWh of electricity.

Boilers: the cheapest capacity in the catalogue

What it is Burn a fuel to make heat directly, with none of the turbine machinery in between. Because nothing is being converted into work, the efficiency is close to one, 1.03 for the district boiler in our table.[a]

Cost signature Almost entirely fuel. Boilers are cheap to build, 58 EUR/kW, well under a hundredth of a nuclear plant per unit of capacity, and there is no efficiency advantage to be bought by spending more. That is the peaker signature of Section 3.2 on the heat side: build a lot of it and run it rarely.

What limits deployment Very little. A boiler is the technology a district heating system falls back on, so its constraint is the fuel price and the carbon price rather than anything about the machine.

[a]Fuels can be measured by their *lower* or *higher* heating value, depending on whether the energy in the water vapour from combustion is counted; European sources, this note included, use the lower. A condensing boiler recovers some of that vapour heat, so it delivers slightly more than the lower heating value says went in, and the ratio comes out above one. The heat pump's coefficient of performance below exceeds one for a different reason.

Heat pumps: not an efficiency

What it is A machine that makes heat from electricity, so its running cost is the electricity price. A heat pump's **coefficient of performance**, heat delivered per unit of electricity consumed, is routinely between three and five, and is 3.15 in our table. A heat pump does not make heat, it *moves* heat from outside to inside, and the electricity pays for the moving rather than for the heat itself. An efficiency is the share of an input that survives a conversion and is bounded by one for that reason; here the electricity is not converted into the heat at all, so there is no share and no bound.

The COP is not constant. It falls as the temperature difference the pump has to work across grows, which means it is lowest exactly when it is coldest and heat is most wanted (Staffell et al. 2012). Assuming a constant annual COP therefore flatters the technology.

Cost signature Capital and electricity, in roughly equal parts at our assumptions. The COP plays the role of η in Equation (1), which is why it sits in the same column of Table 3.

What limits deployment A district heating network, or in a single building a set of radiators sized for water that is not very hot. Also the electricity price, and specifically the taxes and tariffs on it: a heat pump competes against a boiler burning untaxed or lightly taxed fuel, so the comparison is often decided by policy rather than by the machines.

Co-generation: one plant, two products

What it is Combined heat and power, CHP: a thermal plant that sells its waste heat instead of throwing it away. Total conversion efficiency rises from perhaps a third to something like 85–90% as a result. It is the reason the biomass and waste plants in our table have such low *electrical* efficiencies: they are built for both products.

Cost signature Capital, and flexibility. A *back-pressure* plant produces heat and electricity in a roughly fixed ratio, so it cannot choose one without the other. *Extraction* plants can trade a little heat for extra power, and *back-pressure with bypass* can trade the other way, at some cost. A co-generation plant therefore does not have a marginal cost of electricity on its own; Section 7.4 returns to this.

What limits deployment A district heating network has to exist first, and be big enough and close enough, since heat does not travel. That is why co-generation is common in

Table 1: Cost components by technology. High and Low are relative to the other rows, not absolute. "Adjustment" means the economic and technical difficulty of delivering a *chosen* output profile, so a technology that cannot raise output on command scores High, which is why solar and nuclear share an entry in that column for different reasons. Every entry varies with geography and vintage. Hand-authored; see Appendix D.

Technology	Investment	Running	Adjustment	Time to build
Solar PV, utility	Medium	Low	High	Low
Solar PV, rooftop	High	Low	High	Low
Onshore wind	Medium	Low	High	Medium
Offshore wind	High	Low	High	High
Hydro, reservoir	High	Low	Low	High
Hydro, run-of-river	High	Low	High	High
Nuclear	Very high	Low	High	Very high
Coal	High	Medium	High	High
Gas CCGT	Medium	Medium	Medium	Medium
Gas OCGT (peaker)	Low	High	Low	Low
Oil peaker	Low	Very high	Low	Low
Biomass CHP	High	Medium	High	Medium
Waste CHP	Very high	High	High	Medium

Table 2: Technology cost estimates, 2025 vintage, in 2020 euros. η is conversion efficiency, shown only where there is a fuel to convert; c is the marginal cost of Equation (1); e is the emission rate of Equation (2). "Duty" is the reference utilisation, an assumption of mine, and LCoE is the levelised cost at that duty and at a 7% real discount rate. Marginal costs and emission rates are computed from the efficiency and the fuel assumptions. A dash in the e column means the plant emits nothing; 0.00* means it emits and is counted as zero by the convention discussed in Section 3.5. Generated from `data/processed/`; provenance in Appendix D.

Technology	Investment EUR/kW	Fixed O&M %/yr	Life yr	η	c EUR/MWh	e t/MWh	Duty h/yr	LCoE EUR/MWh
Solar PV, utility	473	2.2	38	–	0.0	–	1,007	46
Solar PV, rooftop	880	1.3	38	–	0.0	–	876	89
Onshore wind	1,140	1.2	28	–	1.5	–	2,891	39
Offshore wind	1,884	2.4	30	–	0.0	–	4,380	45
Hydro, reservoir	2,275	1.0	80	–	0.0	–	3,504	52
Hydro, run-of-river	3,412	2.0	80	–	0.0	–	3,942	78
Nuclear	8,594	1.3	40	0.33	14.0	–	7,884	110
Coal	3,827	1.3	40	0.36	37.0	0.94	4,380	114
Gas CCGT	905	3.3	25	0.57	66.0	0.35	3,504	97
Gas OCGT (peaker)	470	1.8	25	0.41	91.2	0.49	438	202
Oil peaker	363	2.5	25	0.35	163.5	0.73	175	393
Biomass CHP	3,642	2.9	25	0.27	97.7	0.00*	5,256	177
Waste CHP	8,830	2.4	25	0.21	28.5	0.73	7,446	158

Denmark and rare in most of Europe, and it is a constraint on the system rather than on the plant.

3.11 The tour in two tables

Table 1 is the qualitative summary: for each technology, whether its investment, running, adjustment and construction costs are high or low relative to the others. It is a judgement rather than a measurement. Table 2 is the same table with money in it, generated from the cost data.

Table 3: The technologies whose product is not electricity. Investment is per kW of power capacity, except for battery cells, where it is per kWh of energy capacity; the two are priced separately and cannot be added. No levelised cost is reported for stores, which produce no energy of their own. The fifth column is a conversion efficiency for the boiler and a coefficient of performance for the heat pump, which play the same role in Equation (1). The heat pump's marginal cost assumes electricity at 60 EUR/MWh.

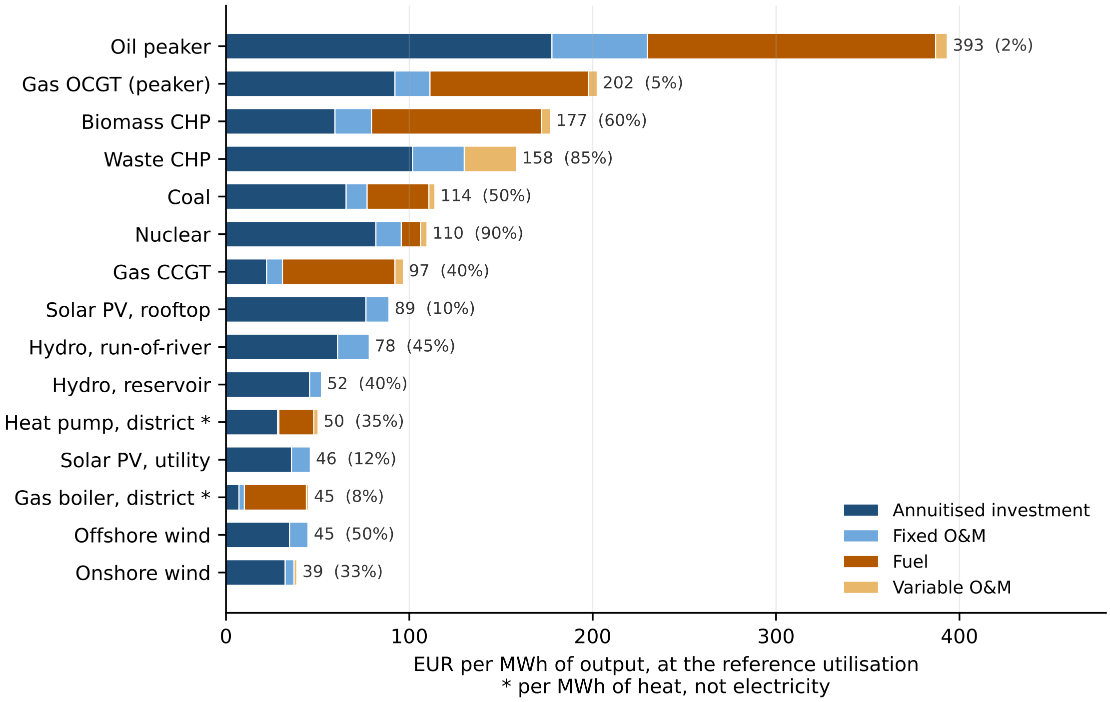
Technology	Product	Investment EUR/kW(h)	Fixed O&M %/yr	Life yr	η / COP	c EUR/MWh
Pumped hydro storage	Electricity	2,275	1.0	80	–	0.0
Battery inverter	Electricity (power)	147	0.2	10	–	0.0
Battery cells	Electricity (energy)	128	0.0	22	–	0.0
Heat pump, district	Heat	1,007	0.2	25	3.15	21.4
Gas boiler, district	Heat	58	3.5	25	1.03	34.9

Figure 3.1 puts the two together: one bar per technology, splitting a euro per MWh at the reference utilisation stated in brackets. Three of the four categories of Section 2.1 are shown, with the variable category split into its fuel and non-fuel halves. **Adjustment costs are absent, and they cannot be added:** a cost of a pattern of operation has no value at a single utilisation, and a single utilisation is all this axis knows about. The oil peaker and the coal plant are the two rows most flattered by the omission. The figure shows the following.

- **The renewables and the peakers are mirror images.** Onshore wind, offshore wind and solar are nearly solid blue: they are capital, and almost nothing else. The oil peaker and the gas peaker are mostly brown: they are fuel. Nothing about the two groups can be compared without saying how much each runs.
- **The cheapest and most expensive rows differ by a factor of ten,** and most of the difference is utilisation rather than the machines: the oil peaker is expensive per MWh because it is assumed to run 2.0% of the year, and it is assumed to run 2.0% of the year because that is what a peaker is for.
- **Nuclear and coal end up close together** for opposite reasons, nuclear being capital spread over a great many hours and coal moderate capital plus a real fuel bill, and neither is competitive here with a combined-cycle plant at these fuel prices.
- **Two rows are not electricity at all.** The heat pump and the gas boiler are marked with an asterisk; their bars are euros per MWh of *heat*, and comparing them with the rest of the figure is a category error.
- **The most expensive plant to build is not the nuclear one** but the waste plant, at 8,830 EUR/kW against nuclear's 8,594. The plant is built to dispose of waste and the electricity is a by-product, so its capital cost per kW of electricity answers no question anyone asks.

Every bar depends on the utilisation in brackets, which is an assumption of mine; change it and the ordering changes. Section 4 does exactly that.

Figure 3.1: Cost per MWh by technology at the reference utilisation shown in brackets, split into the three cost categories of Section 2.1 that have a per-MWh form, with the variable category shown as fuel and non-fuel separately. Adjustment costs are not shown. 2020 euros, a 7% real discount rate, and the fuel prices of Appendix D. Carbon is not included; at 80 EUR per tonne it would add 76 EUR/MWh to coal and 28 to the combined-cycle plant.



3.12 What the tour leaves out

Finally, what is not in the catalogue, and why.

- **Electrolysis and power-to-X.** Covered elsewhere in the course, where the relevant question is about the hydrogen market rather than about the machine.
- **Geothermal, concentrated solar, tidal and wave.** Real technologies with real niches, none of which is a plausible large-scale contributor in northern Europe.
- **Carbon capture.** A modifier on the thermal technologies above rather than a technology of its own, and one whose costs are still dominated by a small number of demonstration projects.
- **Long-duration and seasonal storage.** The gap that batteries do not fill. There is no consensus technology.
- **Demand-side resources.** Not generation at all, though they compete with it; they appear briefly in Section 7.4.

4 Levelised cost

Section 2 and Section 3 described a technology as a *list* of costs. This section collapses the list into a single number, because that is what every report, press release and ministerial briefing does. The collapse is useful and it is lossy, and the section is about what is lost. What happens to costs over time is Section 5; why the same machine costs different amounts in different places is Section 6.

4.1 The definition

The **levelised cost of energy** is the discounted lifetime cost of a plant divided by its discounted lifetime output:

$$\text{LCoE} \;=\; \frac{\displaystyle\sum_{t=0}^{T} \frac{I_t + F_t + V_t}{(1+r)^t}}{\displaystyle\sum_{t=0}^{T} \frac{E_t}{(1+r)^t}}, \tag{4}$$

where I_t, F_t and V_t are investment, fixed and variable costs in year t, E_t is energy produced, r is the discount rate and T the lifetime. The units are EUR per MWh, and it reads as *the constant price at which the plant would exactly break even.* Two remarks on the expression.

First, **the denominator is discounted too**. Discounting energy may look odd, since a MWh in 2040 is the same physical thing as a MWh today, but what is being discounted is the *revenue* it earns: Equation (4) is the price that sets the present value of revenue equal to the present value of cost, so both sides are discounted.

Second, **the discount rate is a financing assumption, not a technical parameter**. It describes what the money costs, not the machine, and Section 4.3 shows that it matters more than most things that do describe the machine.[2]

For a plant with constant annual output, Equation (4) simplifies into the form this note uses:

$$\text{LCoE} \;=\; \underbrace{\frac{I \cdot \big(a(r,T) + f\big)}{8760 \cdot CF}}_{\text{capacity cost, per MWh}} \;+\; \underbrace{c}_{\text{running cost}}, \qquad a(r,T) = \frac{r}{1-(1+r)^{-T}}, \tag{5}$$

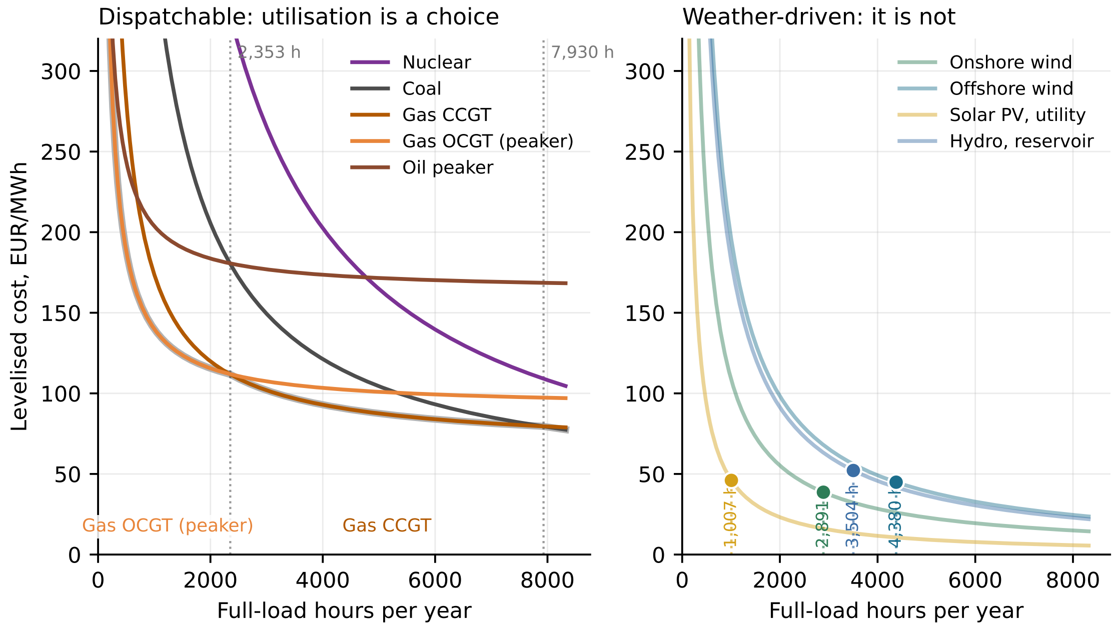
where I is the investment cost per kW, f the fixed O&M rate, c the marginal cost of Equation (1), and $a(r,T)$ the **annuity factor**, the share of the investment that has to be recovered each year (Appendix B). There are two terms. The first falls as the plant runs more; the second does not. Every technology is a different pair of numbers in front of those two terms.

Building versus running

Equation (5) prices a plant that has not been built yet. For a plant that already exists the investment is sunk: the money is gone and has no bearing on any decision from today onwards,

[2] It must be a *real* rate here, because every cost in this note is in constant 2020 euros. Mixing a nominal discount rate with real costs charges the plant for inflation twice, and can raise a levelised cost by half.

Figure 4.1: Levelised cost against utilisation. Left: the thermal plants, whose utilisation is a decision; the thick grey line is the lower envelope and the labels beneath it name the cheapest technology at each duty. Right: the weather-driven technologies, where the curve is the same arithmetic but only one point on it is available, the marker, at the capacity factor the resource delivers. 7% real discount rate, 2020 euros.



and what the plant avoids by not running is c alone. The same technology is therefore governed by two different numbers depending on the question.

- **Whether to build** compares the levelised cost, the whole of Equation (5), against the revenue the plant expects to earn.
- **Whether to run, and whether to keep** compares the marginal cost c against the price. The capacity term is irrelevant to both, except that fixed O&M has to be paid to stay available at all, and so enters the decision to close.

This is why the coal plant in Table 2, whose levelised cost of 114 EUR/MWh makes it look uncompetitive, would run at any price above 37: nobody is proposing to build it. A large part of the existing fleet is in this position, earning too little to have justified its construction and too much to justify closing it. Section 7.2 builds its vocabulary on Equation (5), so that vocabulary describes what a system would build from scratch.

4.2 Levelised cost depends on how much you run

Figure 4.1 plots Equation (5) as a function of the capacity factor.

The left panel shows the following.

- **Every curve is a hyperbola plus a constant.** The constant is the marginal cost, the level the curve flattens towards on the right, and the steepness on the left is the capacity cost. Capital-heavy technologies fall a long way; fuel-heavy ones start lower and barely move.

- **The curves cross.** Below about 131 full-load hours the cheapest way to serve a MWh is the oil peaker, the plant that was cheapest to build. From there to 2,353 hours it is the gas peaker, and above 2,353 hours the combined-cycle plant, because by then there are enough hours to spread its extra capital over.
- **The lower envelope is a portfolio.** No single technology is cheapest everywhere, which is the formal reason a system wants a mix of plants rather than many copies of the cheapest one. A picture of this kind is called a **screening curve**, and its envelope answers the question "what would it be cheapest to *build* for a plant that runs this many hours?". It is the origin of the vocabulary in Section 7.2.
- **Nuclear never appears on the envelope at these assumptions.** At a 7% discount rate and 8,594 EUR/kW it is more expensive than a combined-cycle plant at every duty; Section 4.3 shows why.

The right panel makes the caveat. The arithmetic of Equation (5) applies to wind and solar exactly as it does to a gas turbine, but a wind farm cannot decide to run 6,000 hours a year: the curve exists, and only the marked point on it is available. Reading a levelised cost for a variable technology off an arbitrary utilisation is therefore meaningless, and quoting one without saying which site is meant is nearly so. In our numbers, onshore wind costs 51 EUR/MWh at a poor site (25%) and 29 at a very good one (45%). That is the same machine at the same price; the entire difference is where you put it.

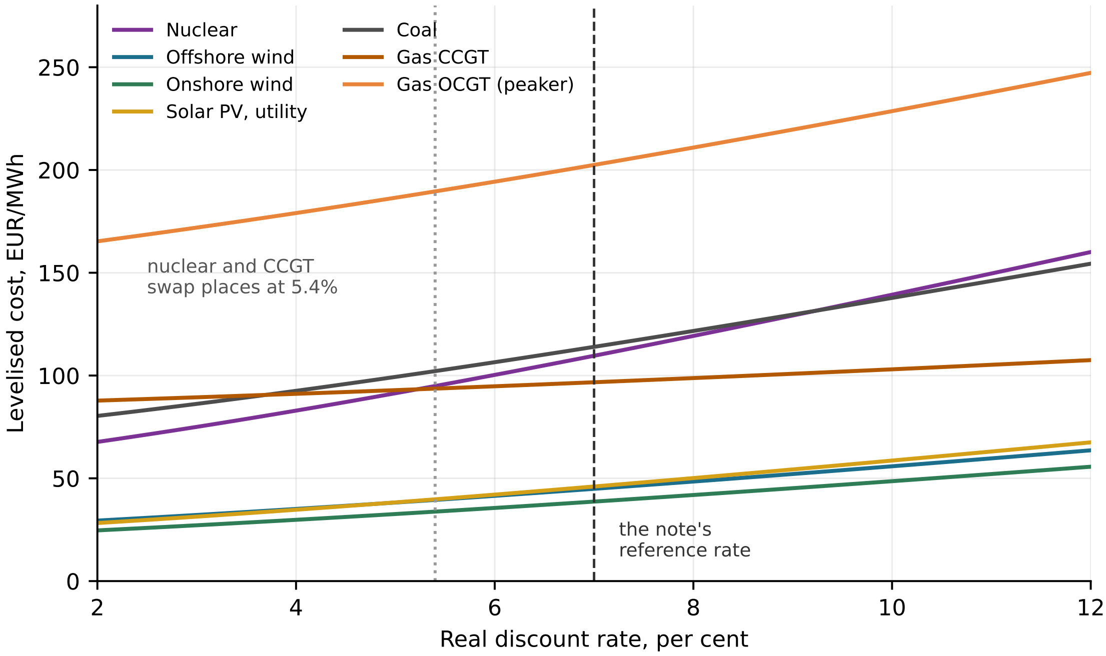
4.3 The cost of capital

For a technology with no fuel cost and a thirty-year life, the levelised cost is largely a statement about financing rather than about engineering. The rate that belongs in Equation (5) is the project's **weighted average cost of capital**, or WACC: the return that the debt and the equity together require, weighted by how much of each is used. It is the r of Equation (5), a property of the financing rather than of the plant, and it differs across technologies, countries and policy regimes far more than the machines do.

Figure 4.2 sweeps the rate from 2% to 12%, holding everything else fixed.

- **The lines have very different slopes, and the slope is the capital share.** The gas peaker barely moves; nuclear more than doubles across the range, from 75 EUR/MWh at 3% to 150 at 11%.
- **The ranking is not preserved.** Nuclear and the combined-cycle plant swap places at 5.4%. Below that rate nuclear is the cheaper way to serve baseload; above it, gas is. An assumption that appears in no engineering document decides the answer.
- **The renewables move a lot in proportion and little in level,** because they start low: offshore wind goes from 32 to 60 EUR/MWh, roughly doubling, but stays below everything thermal throughout.

Figure 4.2: Levelised cost against the real discount rate, each technology at its own reference utilisation. Nothing about the machines changes along the horizontal axis; only what the money costs.



Required returns do differ across technologies, countries and policy regimes, and by a lot: Steffen (2020) finds spreads of several percentage points across European renewable projects alone, and Egli et al. (2019) show that energy system models applying one uniform cost of capital to every technology produce systematically biased capacity mixes.

The same arithmetic explains a good deal of renewables policy. A contract for difference, a feed-in tariff or a long-term power purchase agreement does not change what a turbine costs. It removes revenue risk, which lowers the required return, which by Equation (5) lowers the cost of the energy. In levelised-cost terms, much of renewables policy is a discount-rate intervention.

4.4 Estimates, and why published figures disagree

Published levelised cost estimates for the same technology in the same year routinely differ by a factor of two, and the difference is mostly not measurement error. It decomposes into four things.

- **Resource quality.** The same wind turbine costs 51 or 29 EUR/MWh depending only on where it stands. Solar in Denmark and solar in Spain are different numbers for identical panels: 53 against 21 in our arithmetic.
- **The cost of capital**, per Section 4.3. A study using 4% and a study using 9% are not disagreeing about the technology.
- **Labour, permitting and supply chains**, which differ enough across countries that the same plant costs different amounts to build.

- **The boundary of the estimate.** Does the figure include the grid connection? The reinforcement of the network further back? Decommissioning? Land? Insurance? Each choice is defensible, and two defensible choices can differ by 30%.

A comparison assembled from sources with different boundaries, different currency years and different capital costs is therefore not a comparison of technologies but of methodologies. When handed a levelised cost, ask four questions — what discount rate, what capacity factor, what currency year, what is inside the boundary — and if they cannot be answered, the number's meaning is unknown. Every number in this note is answerable on all four counts, which is what Appendix D is for. Section 6 compares regions by importing their *inputs* and computing the costs through Equation (5), so that both sides of every comparison come out of the same equation.

The questions applied to this note. As Section 1.3 noted, roughly three fifths of the parameters in Table 2 come from a Danish catalogue, and the discount rate, the fuel prices and every reference capacity factor are assumptions listed in Appendix D. Two consequences follow. The levels in this note are *north-west European* levels, so the comparison between two rows is more reliable than either row's value. And the boundary is the same for every row, which is what makes those comparisons legitimate.

4.5 What the levelised cost does not tell you

The levelised cost is an *average cost per MWh produced.* It contains no information about *when* or *where* those MWh arrive, and for electricity that omission is serious: a MWh at eight on a still January evening and a MWh at noon on a windy June Sunday are not the same good, are not worth the same, and may not have the same sign.

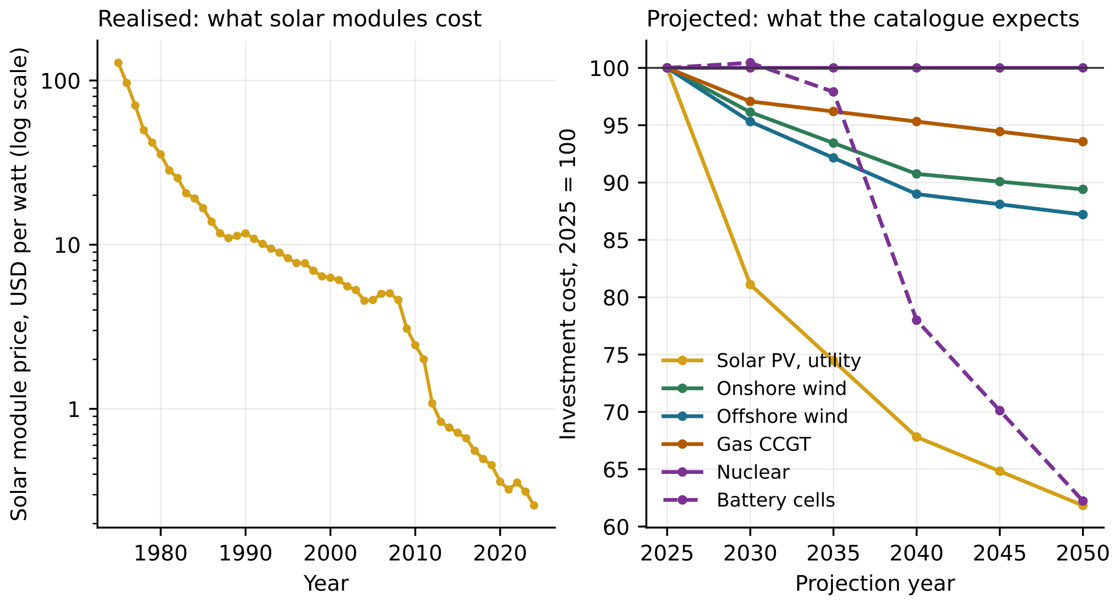
So two technologies with the same levelised cost are not equally useful to a system, and a system built to minimise levelised cost is not minimising cost. The literature calls the difference **integration costs** and splits it into three parts: *profile* costs, from producing at the wrong times; *balancing* costs, from not being predictable; and *grid-related* costs, from producing in the wrong places (Hirth et al. 2015). Heptonstall and Gross (2021) review what has been measured, and Joskow (2011) sets out why comparing intermittent and dispatchable technologies on levelised cost is a category error in the first place.

This note goes no further. The right way to handle it is to compare the *value* of what each technology produces, hour by hour, inside a model of the system, which is a different set of tools. What to take from this one is the negative result: the levelised cost is a description of a plant's cost structure, not a verdict on whether to build it.

5 Costs over time: projections and learning

Every cost in Section 4 was a 2025 figure. This section asks what happens to them over time, in two ways: forwards, by looking at what published projections say and how such projections have

Figure 5.1: Left: the realised world average price of solar photovoltaic modules, 1975–2024, on a log scale (Ritchie et al. 2024). Right: the investment costs the pinned catalogue projects for 2025–2050, indexed to 2025 = 100 (PyPSA, Danish Energy Agency, et al. 2025). The two panels are drawn to very different vertical scales.



fared before (Section 5.1); and backwards, by asking why costs fall at all (Section 5.2).

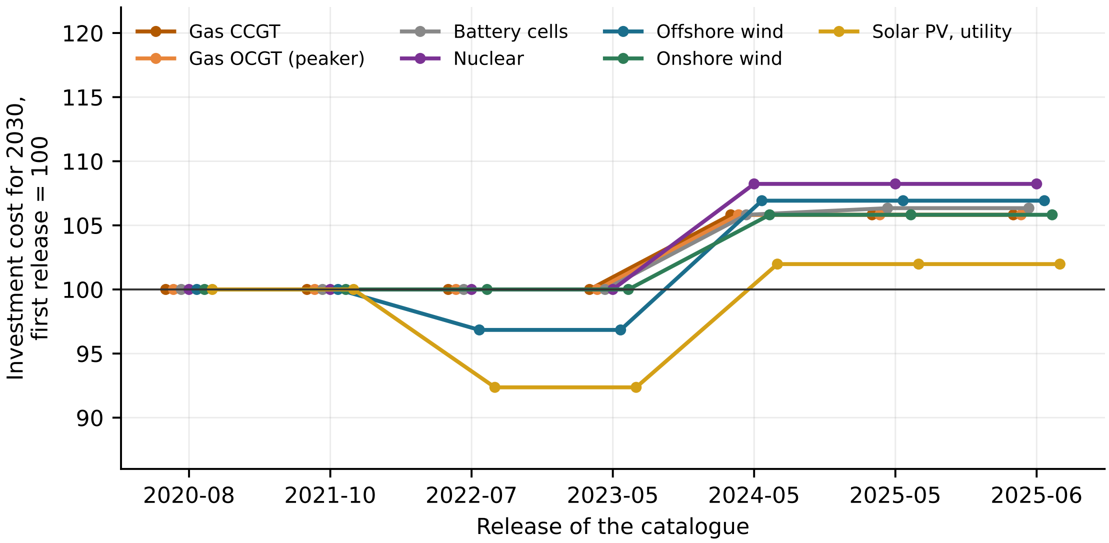
5.1 Projections, and their track record

Every number in Table 2 is a 2025 estimate, and every energy system study that matters is about 2040 or 2050. So the estimates have to be projected forward. Figure 5.1 puts the two halves of the question side by side: what has happened, and what the catalogue expects.

- **The realised history is extraordinary.** Solar modules cost about 128 USD per watt in 1975 and about 0.26 in 2024, a fall of more than two orders of magnitude, drawn on a log scale because it does not fit on a linear one.
- **The projections are cautious by comparison.** The catalogue expects utility solar to reach 62 per cent of its 2025 investment cost by 2050 and battery cells 62 per cent, while onshore wind gets to 89 and nuclear and hydro do not move at all.
- **Some technologies are projected to be flat because nobody expects them to change,** and some because no one has a defensible basis for saying they will. Those are different statements and the data does not distinguish them.

Cost assumptions are the largest single source of uncertainty in any long-horizon statement about the energy system, larger often than the weather year or the demand projection, and a projection made in 2010 is evidence about 2010.

Figure 5.2: The same projection, as successive releases of the same catalogue stated it: investment cost for 2030, indexed to the first release. Series that lie on top of one another were not revised. Money as published in each release, which is what makes the step in 2024 visible.



How much does a catalogue actually move?

A smaller version of the same question: if a study is built on a published cost catalogue, how much does the answer depend on which release was downloaded? Figure 5.2 tracks the *same* projection, investment cost for 2030, across successive annual releases of the catalogue used in this note.

The answer is: less than one would expect. Between 2020 and 2025 the projected 2030 cost of onshore wind moved by +6 per cent, nuclear by +8 and utility solar by +2. Most of the movement is the step between the 2023 and 2024 releases, and most of *that* is a change of currency year rather than a change of view about the technology. Over the same five years, actual solar module prices roughly halved.

Two lessons. Published catalogues are updated far less often than the world moves, and when they are, a change in the number does not always mean a change in the estimate. Always check the currency year before concluding that something got more expensive.

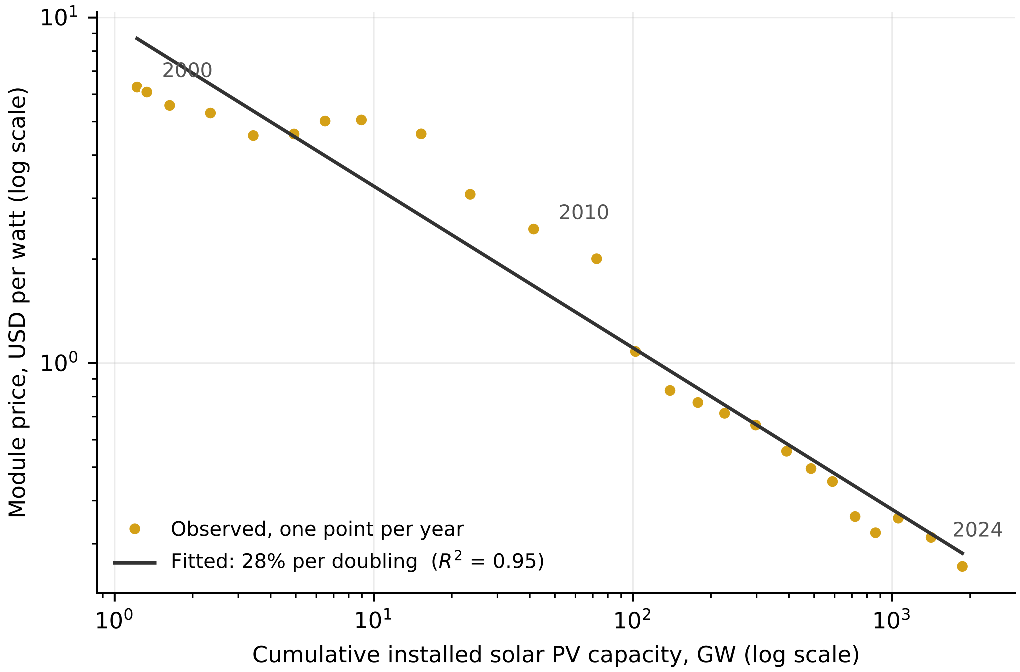
5.2 Learning curves

Why do costs fall at all? The standard empirical answer is the **learning curve** (or experience curve): cost declines as a power function of *cumulative* production or installed capacity, so that each doubling of cumulative deployment cuts cost by a constant percentage. That percentage is the **learning rate**. In logs the relationship is linear, which is why these figures are drawn log–log.

Figure 5.3 fits it to solar modules over 2000–2024, the period for which we have both a price series and a capacity series.

- **The fit is remarkably good.** Over 10.6 doublings of cumulative capacity, from 1.2 GW to 1,866 GW, the module price fell from 6.29 to 0.26 USD per watt, and a single straight line

Figure 5.3: The experience curve for solar photovoltaic modules, 2000–2024. One point per year; both axes logarithmic. Data from Ritchie et al. (2024); the fit is an ordinary least squares regression of log price on log cumulative capacity.



accounts for $R^2 = 0.95$ of it.

- **The implied learning rate is 28% per doubling** over this window. That is at the high end of the published range; Rubin et al. (2015) survey learning rates across electricity supply technologies and report solar PV in the low twenties, with most other technologies well below that.
- **The relationship is not universal.** It holds for things that are manufactured in factories in large numbers. It fails for things built one at a time on unique sites: hydro has no meaningful learning rate, and Grubler (2010) finds the French nuclear programme getting *more* expensive as it built more reactors.

What the fitted line hides

A straight line through twenty-five points is easy to draw and hard to interpret, and the difficulties are economic rather than statistical.

Cumulative capacity is not exogenous. Deployment happened because costs were falling and because policy paid for it; costs fell partly because deployment happened. A regression of one on the other recovers a correlation whose causal content is unclear. Nemet (2006) takes the solar case apart into its physical drivers, plant scale, module efficiency and the silicon price, and finds that a good deal of what the learning curve attributes to experience is scale and input prices.

Deployment and research are different things. One-factor curves like the one in Figure 5.3

explain cost by deployment alone. Two-factor curves separate deployment from accumulated R&D, and generally attribute less to the former.

A fitted rate extrapolated forward is a forecast. Section 5.1 showed how those go, although Way et al. (2022) argue the errors have been systematically in one direction, with the conventional models consistently too pessimistic about exactly the technologies with high learning rates.

One policy question follows. If learning is a knowledge spillover — if my building a solar farm makes your solar farm cheaper, and I cannot charge you for that — then deployment subsidies are correcting an externality, and the case for them is a standard one. If the correlation is mostly scale economies and falling input prices, that case is much weaker. The evidence does not settle it, and the argument is one for the lectures.

One consequence for modelling: if costs depend on cumulative deployment, the planner's problem stops being convex, and the machinery of linear programming stops applying. That is a substantial part of why energy system models take technology costs as given, as this note does, and then test how much the answer depended on them.

6 Costs across geography

Section 5 asked what happens to a cost over time. This section asks why the same machine, bought from the same factory in the same year, costs a different amount to run in Denmark, in Texas and in Jiangsu. Statements such as "renewables are cheaper in America" or "China builds nuclear for a fraction of what Europe pays" are broadly true and say nothing about which of several different things is responsible, and the policy conclusion differs in each case.

6.1 Method

Comparing a published levelised cost for Europe with one for the United States does not work: the two will have been produced with different boundaries, different currency years, different capacity factors and different costs of capital, and the difference between them is an unknown mixture of the world and the method (Section 4.4).

Instead, we use Equation (5) directly. It takes an investment cost, a lifetime and fixed O&M, a capacity factor, a marginal cost and a discount rate, so we import the *inputs* for each region and compute the levelised cost ourselves, at our own boundary and in our own currency year. Table 4 does this. The European column is the note so far: the numbers of Table 2, unchanged. The other two columns change four inputs and nothing else, so every level in this section comes out of the same arithmetic.

6.2 The five channels

A cross-country cost gap decomposes into five channels: the four questions of Section 4.4 asked across space rather than across studies, and policy.

Table 4: The same technologies under different regional inputs. The upper block is inputs, **all of which are illustrative assumptions of mine rather than measurements**: round numbers, right in direction and rough magnitude, with the reasoning for each in Appendix D. The lower block is computed from those inputs by Equation (5), using this note's investment costs, lifetimes and fixed O&M throughout. The European column reproduces Table 2. The lower block is not an estimate of what electricity costs in these places; it is what each channel is worth.

Input or result	Europe	United States	China
Inputs — assumptions, see Appendix D			
Real discount rate	7.0%	6.0%	4.5%
Solar capacity factor	11.5%	24.0%	17.0%
Solar investment (Europe = 1)	1.00	1.15	0.65
Nuclear investment (Europe = 1)	1.00	1.15	0.35
Gas price, EUR/MWh$_{\mathrm{th}}$	35	12	40
Carbon price, EUR/tCO_2	80	0	10
Computed here from Equation (5), EUR/MWh			
Utility solar	46	23	16
Nuclear	110	113	40
Gas CCGT	97	54	101
Gas CCGT, marginal cost	66	26	75
with carbon	94	26	78

Resource. The sun and the wind are not distributed evenly, and Section 4.2 showed what that does: the same panel earns its capital back over 11.5% of the year in Denmark and 24.0% in the American south-west. This channel is not a cost difference. There is more sun.

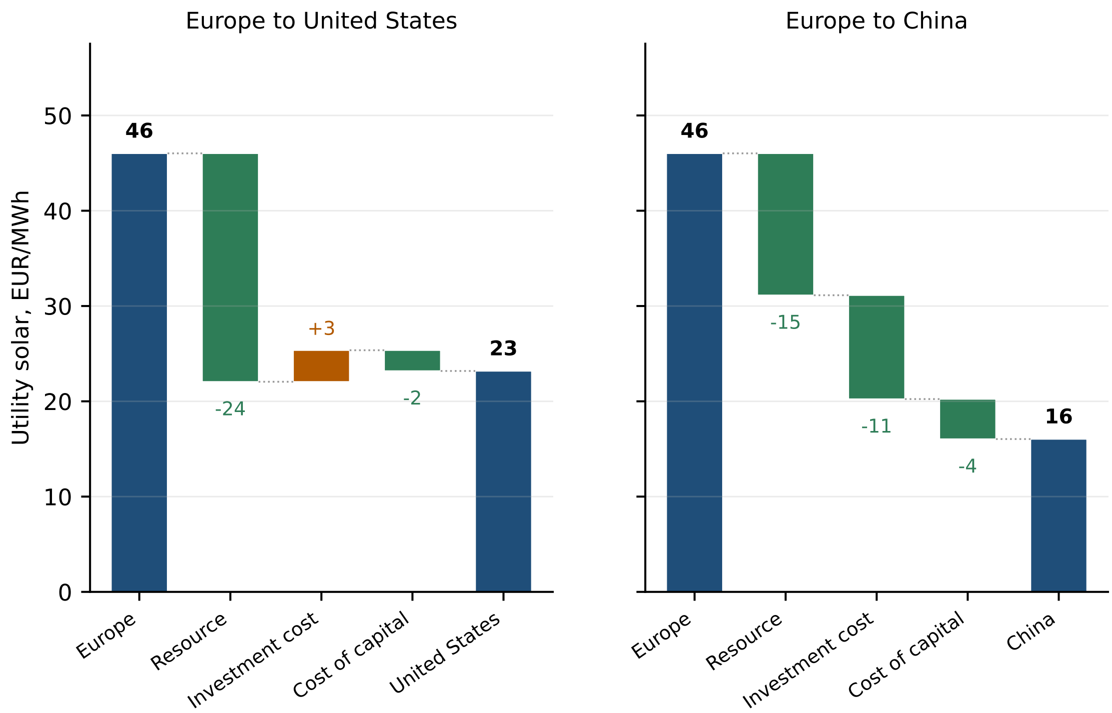
Cost of capital. Section 4.3 showed that the discount rate moves a capital-intensive technology more than most of its engineering does. Required returns differ across countries by several percentage points, for reasons that are about currency risk, policy stability and who is lending rather than about the machine.

Investment cost. A solar module is a manufactured good made in a small number of very large factories and shipped everywhere, so it has something close to a *world price.* What surrounds it does not: the mounting steel, the trenching, the electrician, the permit, the interconnection study, the land lease and the sales tax are bought locally and priced locally. A technology's investment cost therefore splits into a **traded** component that converges across countries and a **non-traded** component that does not, and as modules have got cheaper, the non-traded share has grown. *The geography of cost is mostly the geography of the non-traded part.* That is why identical panels can be installed for very different sums in two rich countries, and why the gap is largest for the smallest installations, where the electrician is the largest line in the bill.

Fuel. Coal and oil are traded globally and cost roughly the same everywhere. Gas moves by pipeline or as a liquid at considerable expense, so regional gas markets can diverge for years at a time. Since 2022 European gas has traded at a persistent multiple of American gas, and the consequence for a gas plant's marginal cost is immediate.

Policy. A carbon price, a tax credit or a contract for difference changes what a plant costs its owner without changing what it costs the world to build or run. This channel is different in kind

Figure 6.1: From the European levelised cost of utility solar to the American and Chinese ones, one input at a time. Blue bars are levels; green and brown bars are the change made by a single input, holding the others at the value they have reached. The steps do not commute, since the capacity factor divides the capital term that the discount rate scales, so the split between channels depends on the order they are applied in: resource, then investment cost, then cost of capital. The totals do not depend on the order. 2020 euros.



from the other four.

6.3 From a European cost to an American one

Figure 6.1 takes the note's European utility-solar levelised cost and walks it to each of the other two columns of Table 4, changing one input at a time.

Three things stand out.

- **The American number is lower than the European one despite the plant being more expensive to build.** Investment cost *adds* +3 EUR/MWh going west. The resource contributes −24 and the cost of capital a further −2, and between them they more than swamp it. Solar is cheaper in the United States than in Denmark because of the sun, and for no other reason in this arithmetic.
- **The Chinese gap is a cost gap, and it is split.** The resource contributes −15, the investment cost −11 and the cost of capital −4. The modules are the same modules, at much the same world price; what is cheaper is everything around them, plus the money.
- **No single channel dominates everywhere.** The resource is the largest single channel in both panels, but it accounts for essentially all of the American gap and only about half of the Chinese one.

6.4 Two earlier claims, qualified

"Nuclear never appears on the envelope." Section 4.2 found that at a 7% discount rate and 8,594 EUR/kW, nuclear is more expensive than a combined-cycle plant at every duty. That is a statement about *Western construction experience at a European cost of capital* rather than about reactors. Almost all of Table 2 is Danish in origin, but the nuclear row comes, through the compilation described in Appendix D, from an American levelised-cost analysis, and so carries the recent Western record of first-of-a-kind projects that have overrun heavily. Under the Chinese column of Table 4, serial construction of a repeated design financed at an administered rate, the same equation returns 40 EUR/MWh instead of 110, which would put nuclear on the envelope comfortably.

The accounting behind the Chinese figure is not directly comparable and the index is the roughest assumption in Table 4, so the level is not one to quote. What it establishes is that the conclusion that nuclear is uncompetitive is a joint statement about construction cost and the cost of capital, both of which a country can be good or bad at, and not a conclusion about the technology.

"The combined-cycle plant is the cheapest thermal option." That was said at a European gas price of 35 EUR/$\mathrm{MWh_{th}}$ and a carbon price of 80 EUR per tonne. At the American gas price the same plant has a marginal cost of 26 EUR/MWh rather than 66, with no carbon price added. A gas plant is therefore a different economic object in the two places, built from identical components, and the merit order, which is decided by exactly these marginal costs, is not a fact about the machines.

6.5 Limitations

The numbers in Table 4 are not measurements. Every input in the upper block is a round figure chosen to be right in direction and in rough magnitude. They size the channels; they do not price electricity in three countries. Appendix D gives the reasoning for each, and a number to quote should come from a source that measures.

The comparison is between rows, not down columns. That the American solar figure is below the European one is robust to almost any reasonable choice of inputs, because it is driven by a resource difference that is not in dispute. That it is below it by a specific amount is not robust at all.

Cheaper is not the same as better sited. A country with abundant sun and cheap capital will show low costs for solar whatever it does. Whether solar is the right thing for that country's system depends on when the sun shines relative to when the demand is, which is the question Section 4.5 left to a model of the system.

The industrial-policy argument. Much of the Chinese investment-cost advantage is in manufacturing, which connects to the learning curve of Section 5.2: a country that builds a great deal of something accumulates the experience that makes it cheap, and whether that is a spillover worth subsidising is the open question that section ended on. This note takes no position on it.

7 Plant types

The previous sections describe machines. This one describes the vocabulary used to talk about their *roles*, and places the technologies of Section 3 within it. It says which plants are conventionally used for what, and why their cost structure makes that natural; what any of them is worth to a system requires a model of the system.

7.1 Dispatchable and variable

A **dispatchable** plant produces what it is told to produce. A **variable**, or **intermittent**, plant produces what its primary energy source offers. Two qualifications follow from the cost taxonomy.

It is a spectrum, not a dichotomy. "Dispatchable" is shorthand for "low adjustment costs", and thermal plants differ enormously in how fast and how cheaply they can move. A gas peaker goes from cold to full output in minutes. A coal plant takes many hours and pays for it in fuel and wear. Both are called dispatchable, and in a system that needs its plants to move around several times a day, the difference between them matters more than the label.

Dispatchable is not the same as thermal. Reservoir hydro is dispatchable and burns nothing; a battery is dispatchable in both directions. Conversely not everything thermal is comfortably dispatchable: nuclear is technically capable and economically disinclined, because with a marginal cost of 14 EUR/MWh there is almost nothing to save by turning down.

Three properties of variable generation

Variable generation is characterised by three properties, which call for different responses.

i. **Temporally variable.** Output changes over the day and over the year, and the pattern is a property of the resource.
ii. **Spatially variable.** Output depends on where the plant stands, which is why Section 4.2 insisted that a levelised cost for wind without a site is not a number.
iii. **Uncertain.** Output cannot be forecast perfectly, even a few hours ahead.

Variability is about the output moving; *uncertainty* is about not knowing in advance that it will. A perfectly predictable but wildly varying output is a scheduling problem, addressed with a portfolio of plants and with storage. An unpredictable one is a reserve problem, addressed with plants held back and with better forecasting.

7.2 Baseload, mid-merit and peaker

The classical partition of a dispatchable fleet is by how much each plant runs, and it is Figure 4.1 read back onto the horizontal axis. That figure is a screening curve: it asks which technology it would be cheapest to *build* for a given number of hours of duty. As Section 4.1 noted, that is not the same question as which plant will run for those hours in a fleet that already exists, since an existing plant answers to its marginal cost alone. The two coincide when a fleet has been built for the world it operates in. The vocabulary below is built on the first question and used every day about the second.

- A **baseload** plant is expensive to build and cheap to run, so it is worth running nearly all the time. It sits on the lower envelope at high full-load hours.
- A **peaker** is the reverse, cheap to build and expensive to run, and earns its place by being available for a few hundred hours a year. In our numbers the oil peaker is cheapest below about 131 full-load hours and the gas peaker from there to 2,353.
- **Mid-merit** is everything between: the combined-cycle plant, which takes over above 2,353 hours.

The defining variable is full-load hours, and Table 5 gives a reference figure for every technology in the catalogue.

One caveat. This vocabulary was built for a system in which demand was the only thing that varied, and the plants were sized against it. Where a large share of generation is variable, the dispatchable fleet is sized against demand *net of* wind and solar, which is a much spikier and less predictable quantity than demand itself. The categories survive, but the duties attached to them shift, generally towards fewer hours and more starts. Working out where they shift to is a modelling exercise.

7.3 Storage and prosumers

A store is neither generation nor demand: it is both, at different times. It buys electricity and sells the same electricity back later, so it appears on both sides of the market, and its cost of acting now is entirely an opportunity cost, what it gives up by not waiting.

The variable that decides what a store is good for is **duration**, the ratio of its energy capacity to its power capacity. A four-hour battery is a tool for the daily cycle: charge at midday when solar is abundant, discharge in the evening. It is not a tool for a two-week wind drought in January, and no reduction in the cost of four-hour batteries turns it into one. Reservoir hydro, whose duration is measured in months, is the only large store in the catalogue that addresses the seasonal problem, and it cannot be built where there are no mountains.

Prosumers. Rooftop solar with a home battery is the same arrangement placed behind the meter. What changes is the accounting: electricity generated and consumed behind the meter avoids not only the wholesale price but also the network tariffs and levies attached to a delivered kWh. Most of a network's cost does not vary with how much energy travels over it, but most network tariffs are charged per kWh delivered, so a household that generates behind its own meter avoids a charge that was recovering a cost its connection still imposes. The economics of a prosumer are therefore substantially a question about tariff design rather than about the technology, which is why the topic sits in a different part of the course.

7.4 Co-generation and coupled markets

Co-generation is a plant type rather than a machine. What defines it is that its output is a *bundle* sold into two markets at once, so whether it is worth running depends on both prices together. A back-pressure plant produces heat and electricity in a roughly fixed ratio and cannot choose one without the other; that constraint, not the boiler, is what makes it a distinct type.

Such a plant does not have a marginal cost of electricity in the ordinary sense. What it gives up to produce a MWh of electricity depends on what the heat is worth at the same moment, so its position in a merit order moves with the heat price, and CHP requires its own treatment in any model that includes it.

Heat pumps are the mirror image: a heat plant whose fuel is electricity, and therefore a piece of **flexible demand**. In a heat system with storage it can choose when to consume, which makes it a participant in the electricity market rather than a passive load. It can only push in one direction, consuming more when electricity is cheap but never supplying when electricity is scarce.

7.5 The mapping, tabulated

Table 5 collects every technology in the catalogue with its controllability, the share of its cost that is capacity rather than output, a reference duty, and the role it is conventionally put to.

Three patterns stand out.

- **The variable technologies are at or near 100% capital.** Their entire cost is incurred before they produce anything, which is why their economics is so sensitive to the discount rate and to the site, and why nothing about how often they run is a decision.
- **Capital share tracks the role, but not monotonically.** Nuclear is 87% capital because its capital is enormous; the gas peaker is 55% capital, despite being the cheapest plant in the table to build, because it runs so rarely that even a small capital cost is spread very thin. The peaker is capital-light per kW and capital-heavy per MWh.
- **The combined-cycle plant is the outlier at 32%,** which is what a mid-merit technology looks like: enough hours to dilute the capital, and a real fuel bill that does not dilute at all.

7.6 What a technology catalogue leaves out

Finally, what a catalogue leaves out.

Table 5: Technologies, cost structure and conventional role. "Capital share" is the share of the levelised cost that is investment plus fixed O&M at the reference duty, so a high number means a plant whose cost is mostly incurred by existing. Typical duty is a reference assumption of mine, not a model result; see Appendix D. Stores are not assigned a capital share, since they have no levelised cost to take a share of. The heat pump's "running cost" is electricity, which is itself made by plants that are mostly capital, so its share describes this machine's own accounts, not the system's.

Technology	Controllability	Capital share % of cost	Typical duty h/yr	Conventional role
Solar PV, utility	Variable	100	1,007	Variable
Solar PV, rooftop	Variable	100	876	Variable
Onshore wind	Variable	96	2,891	Variable
Offshore wind	Variable	100	4,380	Variable
Hydro, reservoir	Dispatchable	100	3,504	Mid-merit / flexible
Hydro, run-of-river	Variable	100	3,942	Variable
Pumped hydro storage	Store	–	1,051	Storage
Nuclear	Dispatchable	87	7,884	Baseload
Coal	Dispatchable	68	4,380	Baseload
Gas CCGT	Dispatchable	32	3,504	Mid-merit
Gas OCGT (peaker)	Dispatchable	55	438	Peaker
Oil peaker	Dispatchable	58	175	Peaker
Biomass CHP	Dispatchable	45	5,256	Baseload / heat
Waste CHP	Dispatchable	82	7,446	Baseload / heat
Battery inverter	Store	–	876	Storage
Battery cells	Store	–	876	Storage
Heat pump, district	Demand-side	57	3,066	Heat, flexible demand
Gas boiler, district	Dispatchable	22	701	Heat, peaking

- **Services other than energy.** A power system needs frequency and voltage held within limits, needs enough inertia to survive a sudden loss of generation, and needs some plants capable of restarting a dead grid. None of this is priced per MWh, and none of it appears in any column of Table 2. It is one of the substantive differences between a synchronous machine and an inverter.
- **Where the plant is.** The grid reinforcement a plant implies depends on its location, and can be a large fraction of its cost. A catalogue indexed by technology cannot express it.
- **Land and acceptance.** Increasingly the binding constraint on onshore wind and utility solar in Europe, and absent from a cost per kW.
- **Supply chains and materials.** The catalogue assumes the machine can be bought. For several technologies that has recently not been true at any price.
- **Decommissioning and waste.** Real costs, incurred long after the revenue stops, and therefore heavily discounted in exactly the cases where they are largest.

A catalogue is a starting point, assembled from central estimates for particular places at particular times. Everything built on it inherits both its structure and its silences.

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Appendices

A Units, and how to avoid the usual mistakes

A short reference. The commonest error in reading technology data is a unit error, not an economic one.

Power and energy

Power is a rate: watts (W), kilowatts (kW = 10^3 W), megawatts (MW = 10^6 W), gigawatts (GW = 10^9 W). A plant's capacity is a power.

Energy is power multiplied by time: watt-hours (Wh), and the same prefixes. A plant's output over a period is an energy. One MW running for one hour delivers one MWh; running flat out for a non-leap year it delivers 8,760 MWh.

The older literature, and some of the earlier material in this course, uses joules instead. The conversions:

$$1 \text{ Wh} = 3600 \text{ J}, \qquad 1 \text{ MWh} = 3.6 \text{ GJ}, \qquad 1 \text{ GJ} = 0.2778 \text{ MWh}.$$

A cost quoted in EUR/GJ is therefore about 3.6 times smaller than the same cost in EUR/MWh, which is a large enough factor that the mistake is usually obvious — but only if you are looking for it.

Thermal and electrical energy

A plant that burns fuel takes in thermal energy and puts out electrical energy, and there is always more of the former. Where the distinction matters this note writes MWh_{th} and MWh_{e}. The efficiency η is the exchange rate between them:

$$1 \text{ MWh}_{\text{e}} \ = \ \frac{1}{\eta} \text{ MWh}_{\text{th}} \quad \text{of fuel.}$$

Fuel prices and carbon contents are quoted per MWh_{th}; marginal costs and emission rates per MWh_{e}; Equation (1) and Equation (2) do the conversion.

A related trap: fuels can be measured by their *lower* or *higher* heating value, depending on whether the energy in the water vapour produced by combustion is counted. The two differ by a few per cent for coal and around 10% for natural gas, and mixing them shifts an efficiency by enough to matter. European sources conventionally use lower heating value, as does this note. That convention is also why the district boiler in Table 3 is credited with an efficiency of 1.03: a condensing boiler recovers some of the vapour heat that the lower heating value does not count, so the ratio comes out above one without anything having gone wrong.

Costs

Published cost data arrives in a small number of standard shapes:

- **Investment** in EUR per kW of capacity — except for the energy component of a store, which is EUR per kWh.
- **Fixed O&M** either in EUR per kW per year, or as a percentage of the investment cost per year. This note's source uses the percentage form, which is why Table 2 reports %/yr.
- **Variable O&M and fuel** in EUR per MWh.
- **Emission factors** in tonnes of CO_2 per MWh_{th}.

Turning these into a single EUR/MWh figure is what Equation (5) does, and it needs one more input that is not in any catalogue: the discount rate.

Capacity factor and full-load hours

$$CF = \frac{\text{energy produced}}{\text{capacity} \times \text{hours}}, \qquad \text{full-load hours} = CF \times 8760.$$

A 40% capacity factor and 3,504 full-load hours are the same statement.

A worked conversion

A combined-cycle plant costs 905 EUR/kW to build, has fixed O&M of 3.3% of the investment per year, an efficiency of 0.57, a variable operating cost of 4.55 EUR/MWh, and a lifetime of 25 years. Gas costs 35 EUR/MWh_{th}. It runs 40% of the year. What does its energy cost?

Step 1: the marginal cost. By Equation (1), $c = 35/0.57 + 4.55 = 66.0$ EUR/MWh.

Step 2: the annual capacity cost. At a 7% discount rate over 25 years the annuity factor is $a = 0.0858$. Adding fixed O&M gives $905 \times (0.0858 + 0.0334) = 108$ EUR per kW per year.

Step 3: spread it over the energy. One kW running 40% of the year produces $8760 \times 0.40/1000 = 3.50$ MWh. So the capacity cost is $108/3.50 = 31$ EUR/MWh.

Step 4: add them. $31 + 66.0 = 97$ EUR/MWh, which is exactly what Table 2 reports for the combined-cycle plant.

Notice how much of the answer is step 2, and how much of step 2 is the discount rate. That is Section 4.3 in miniature.

B Annuities and the levelised cost

This appendix derives the two expressions Section 4 uses and does not prove. Nothing here is difficult; it is put in an appendix so that the main text can state a formula and get on with it.

B.1 The annuity factor

An investment costs I today and the plant lasts T years. We want the constant annual payment A over those T years whose present value is exactly I, discounting at rate r. That is

$$I \;=\; \sum_{t=1}^{T} \frac{A}{(1+r)^t} \;=\; A \cdot \frac{1-(1+r)^{-T}}{r},$$

using the sum of a geometric series. Rearranging,

$$A \;=\; I \cdot a(r,T), \qquad a(r,T) \;=\; \frac{r}{1-(1+r)^{-T}}. \tag{6}$$

The quantity $a(r,T)$ is the **annuity factor**: the share of the investment that has to be recovered each year.

Three checks will fix the intuition.

- As $r \to 0$, $a \to 1/T$: with no time value of money you simply divide the cost by the number of years.
- As $T \to \infty$, $a \to r$: for a perpetual asset you never repay the principal, you only ever pay interest.
- a is always at least r and at least $1/T$, and close to whichever is larger. This is why lifetime stops mattering much once it is long: at 7%, extending a plant's life from 40 years to 80 changes the annuity factor from 0.075 to 0.070, a difference of a few per cent. Extending it from 10 years to 20 changes it from 0.142 to 0.094, a difference of a third.

That last observation is worth keeping. Very long-lived assets are not as valuable as their lifetime suggests, because the last decades are discounted almost to nothing.

B.2 From the definition to the working formula

Equation (4) defined the levelised cost as a ratio of present values. Suppose output is constant at E MWh per year, variable costs are c per MWh, fixed costs are F per year, and the investment I is paid at $t = 0$. Write $D = \sum_{t=1}^{T}(1+r)^{-t}$ for the discount sum. Then

$$\text{LCoE} \;=\; \frac{I + D\,(F + cE)}{D\,E} \;=\; \frac{I}{D\,E} + \frac{F}{E} + c \;=\; \frac{I\,a(r,T) + F}{E} + c,$$

where the last step uses $1/D = a(r,T)$ from Equation (6). Writing F as a rate f on the investment, and $E = 8760 \cdot CF$ per kW of capacity, gives Equation (5) exactly.

So the two forms are the same statement. The ratio-of-present-values definition is the honest one; the annuitised form is the one you can compute in your head, and the one that makes the structure visible.

B.3 Reading the formula

Equation (5) has exactly two terms:

$$\mathrm{LCoE}(CF) \;=\; \underbrace{\frac{K}{8760 \cdot CF}}_{\text{falls with } CF} \;+\; \underbrace{c}_{\text{constant}}\,, \qquad K = I\,\big(a(r,T) + f\big).$$

K is the annual cost per kW of simply having the plant. As a function of the capacity factor, the levelised cost is a rectangular hyperbola with asymptote c: it falls steeply at low utilisation, flattens, and approaches the marginal cost from above without ever reaching it.

Two technologies' curves cross exactly once, at

$$CF^* \;=\; \frac{K_1 - K_2}{8760\,(c_2 - c_1)},$$

where technology 1 has the higher capacity cost and the lower marginal cost. Below CF^* the cheap-to-build plant wins; above it, the cheap-to-run one does. That single expression generates the whole lower envelope of Figure 4.1, and with it the vocabulary of Section 7.2.

B.4 What the measure assumes away

Finally, for completeness, the assumptions buried in Equation (5):

- **Output is constant across years.** Real plants degrade, and solar panels do so predictably, at a few tenths of a per cent per year.
- **Prices are constant.** Fuel prices in particular are not, and a thirty-year gas price is a heroic object.
- **The lifetime is known.** It is an engineering estimate, and plants are frequently retired early for economic or political reasons long before it expires.
- **The discount rate does not depend on the technology.** It does (Steffen 2020; Egli et al. 2019), which is the subject of Section 4.3.
- **Construction is instantaneous.** For a plant that takes a decade to build, interest during construction is a substantial cost that this form does not show.
- **Every MWh is worth the same.** This is the big one, and Section 4.5 is about it.

C Vocabulary

This note is meant to be read once and then used as a reference, so here is everything it defines, in one alphabetical list, with a pointer to where it is discussed properly.

The second half of the list is different. Those are terms this note does *not* develop but that you will certainly meet elsewhere in the course, and which are easy to mistake for something in the first half. They are given one line each so that you have somewhere to look, and no more than that: each of them belongs to a piece of analysis a catalogue cannot do.

C.1 Terms this note uses

Adjustment cost What it costs to *change* output: starting up, ramping, running below minimum load, cycling. A cost of a pattern of operation rather than of a level, which is why it has no per-MWh form. Section 2.1.

Annuity factor $a(r, T) = r/(1 - (1 + r)^{-T})$, the share of an investment that has to be recovered each year. Section B.

Availability The share of nameplate capacity actually available in a given hour, after outages, maintenance or weather. Section 2.2.

Back-pressure plant A co-generation plant that produces heat and electricity in a roughly fixed ratio and cannot choose one without the other. Section 3.10.

Baseload A plant expensive to build and cheap to run, worth running nearly all the time. Section 7.2.

Capacity factor Energy produced divided by energy that would have been produced running flat out. A decision for a dispatchable plant, a fact about the site for a variable one. Section 2.2.

Capital share The share of a levelised cost that is investment plus fixed O&M at a given duty. Section 7.5.

Coefficient of performance (COP) Heat delivered by a heat pump per unit of electricity consumed. Routinely three to five, because the pump moves heat rather than making it, and therefore not an efficiency. Section 3.10.

Combined cycle (CCGT) A gas turbine with a second stage that raises steam from the exhaust. More expensive to build, cheaper to run. Section 3.2.

Co-generation (CHP) A thermal plant that sells its waste heat instead of discarding it, and therefore sells a bundle into two markets. Section 3.10, Section 7.4.

Conversion efficiency (η) The share of a fuel's energy that comes out as electricity. Sits in the denominator of both the marginal cost and the emission rate. Equation (1).

Dispatchable Produces what it is told to produce. Shorthand for "low adjustment costs", and a spectrum rather than a category. Section 7.1.

Duration A store's energy capacity divided by its power capacity. What decides whether a store is a tool for the daily cycle or for something longer. Section 7.3.

Experience curve See *learning curve.*

Full-load hours Capacity factor × 8,760. The same statement in more intuitive units. Section 2.2.

Higher / lower heating value Whether the energy in the water vapour from combustion is counted. European sources, and this note, use the lower. Section A.

Integration cost The system cost of *when* and *where* a technology produces, which a levelised cost omits. Conventionally split into profile, balancing and grid-related costs. Section 4.5.

Intermittent See *variable*.

Learning curve Cost falling as a power function of cumulative deployment, so that each doubling cuts cost by a constant percentage — the *learning rate*. Section 5.2.

Levelised cost of energy (LCoE) Discounted lifetime cost divided by discounted lifetime output: the constant price at which a plant would exactly break even. A statement about building, not about running. Section 4.

Merit order The ranking of plants by marginal cost, in which the cheapest available plant runs first. Because it is a ranking of *marginal* costs, a plant's position in it has nothing to do with what it cost to build. Section 2.1.

Mid-merit Between baseload and peaker: enough hours to dilute the capital, and a real fuel bill. Section 7.2.

Nameplate capacity The output a plant is rated for, as distinct from what is available. Section 2.2.

Open-cycle (OCGT) A gas turbine that lets the exhaust go. Cheap to build, expensive to run, quick to start. Section 3.2.

Peaker A plant cheap to build and expensive to run, earning its place by being available for a few hundred hours a year. Section 7.2.

Primary / secondary energy Energy as found, and energy in a form we have made in order to use it. Section 1.1.

Prosumer A consumer who also generates, usually behind the meter, so that the economics turn on tariff design. Section 7.3.

Round-trip efficiency The share of the electricity put into a store that comes back out. Section 3.9.

Screening curve Levelised cost plotted against duty, whose lower envelope says which technology it would be cheapest to *build* for a given number of hours. Section 4.2.

Sunk cost An investment already made, which is therefore irrelevant to any decision from today onwards — the reason an existing plant answers to its marginal cost alone. Section 4.1.

Traded / non-traded component The part of an investment cost that has a world price (the module, the turbine) and the part bought locally (the labour, the permit, the connection). Section 6.2.

Variable Produces what its primary energy source offers. Characterised by being temporally variable, spatially variable and uncertain — three properties with three different remedies. Section 7.1.

WACC Weighted average cost of capital: the return that debt and equity together require, weighted by how much of each is used. The r of Equation (5). Section 4.3.

C.2 Terms this note does not develop

Each of these requires a model of the system, which is why a catalogue names them and stops. They are collected here so that you can tell them apart from the list above.

Balancing cost The cost of holding reserves against forecast error. One of the three integration costs.

Cannibalisation The tendency of a technology to depress the price in exactly the hours it produces, so that adding more of it lowers what all of it earns. The reason a capture rate falls as deployment grows.

Capacity credit (firm capacity) The amount of conventional capacity a given amount of variable generation can displace while keeping the system as reliable. Considerably less than its nameplate, and falling as more is added — which is why adding wind does not retire gas one-for-one.

Capture price, capture rate The average price a technology actually receives, and its ratio to the average market price. The revenue-side counterpart to a levelised cost, and where *when* you produce enters.

Curtailment Deliberately not producing available output, because the system cannot use it or the price has gone negative.

Levelised cost of storage (LCOS) The storage analogue of a levelised cost. Harder to define than it sounds, since a store's output depends on how it is operated.

Negative prices Prices below zero, which occur when it is cheaper to pay someone to take electricity than to stop producing it. The reason Section 4.5 says two MWh may not have the same sign.

Profile cost The cost of producing at the wrong times. The largest of the three integration costs at high shares of variable generation.

Value factor Capture price divided by the average price; the same idea as a capture rate.

D Provenance: where the numbers come from

This note is largely numbers about machines, so it needs this appendix more than most. Everything below is either a published dataset at a stated version, or an assumption of mine, and the two are kept visibly apart.

D.1 Datasets

Technology costs. PyPSA's `technology-data` (PyPSA, Danish Energy Agency, et al. 2025), fetched at tag v0.13.2, files `outputs/costs_YYYY.csv` for YYYY $\in \{2025, \ldots, 2050\}$. This is a compilation rather than a primary source, and it is worth knowing how it is assembled. The project's documented rule is that **Danish Energy Agency data are preferred** (Danish Energy Agency and Energinet 2024), with other published sources filling the gaps where that catalogue has nothing. Across the parameters this note uses that works out at roughly three fifths DEA, covering everything in the tour except four cases: *coal, lignite and nuclear* come from Lazard's levelised cost analysis, and *hydro, run-of-river and pumped storage* from a DIW Berlin data documentation. Fuel prices come from DIW and the EC's Joint Research Centre, and carbon contents from German federal figures or from stoichiometry. All of it is converted to 2020 euros by the compilation itself. The tag is pinned, so re-running the pipeline reproduces the note's numbers exactly; an unpinned fetch would let them drift silently between builds. It is also how this note stays consistent with the rest of the course material, which reads the same source at the same pin.

Solar module prices and installed capacity. Our World in Data (Ritchie et al. 2024), CC-BY 4.0, series `solar-pv-prices` (world average module price in constant USD per watt, 1975 onwards, compiled from Nemet (2006) and IRENA) and `installed-solar-PV-capacity` (world cumulative installed capacity, 2000 onwards). These are the two series behind Figure 5.3 and the left panel of Figure 5.1. The learning fit uses only the years in which both are available.

Cost vintages. Figure 5.2 takes the same file, `outputs/costs_2030.csv`, from seven successive releases of `technology-data` between August 2020 and June 2025. Money is as published in each release, which is what makes the 2024 step visible.

D.2 Assumptions

The following are mine. None of them is measured, and each is chosen to be plausible for a north-west European system rather than to be right for any particular project.

The discount rate is 7% real, applied uniformly. This is a conventional central value and it is also, as Section 4.3 argues at some length, the assumption most likely to be wrong in an interesting way. Figure 4.2 exists so that you can substitute your own.

Fuel prices are stylised 2025 forward levels , not the dataset's own. Table 6 prints both, so the size and direction of every change is visible. The dataset's fuel prices are inherited from a study now over a decade old and sit well below what European plants have paid recently; the values used here are round numbers at roughly current forward levels, chosen for consistency with the rest of the course.

Battery investment costs are capped at a market level , the one place where this note overrides the dataset's investment numbers. The dataset's battery trajectory has fallen well behind the world market: its 2040 vintage costs roughly what turnkey systems sold for in 2025. Every vintage's four-hour system cost is therefore capped at 165 €/kWh — BloombergNEF's 2025 European average turnkey price (177 $/kWh in the *Energy Storage System Cost Survey 2025*, December 2025, press-release figures; converted at ~1.07 USD/EUR; the global average was 117 $/kWh, the Chinese 73 $/kWh). The split between inverter (€/kW) and cells (€/kWh) keeps the dataset's own ratio, and vintages already below the cap are untouched, so the cap binds for the 2025 and 2030 vintages only. The currency conversion, and reading a nominal-2025 dollar price against the dataset's EUR-2020 basis, are part of the assumption; the processed data records the dataset's value beside the one used, in the same visible-override format as the fuel table. This is, incidentally, the treatment Section 4.4 argues any published catalogue should get: when a source falls behind the market, override it visibly.

Two carbon entries are accounting choices rather than measurements, and the table marks both. Biogenic carbon is counted as zero by convention. Municipal waste carries 0.15 tCO_2 per MWh_{th} — its fossil, largely plastic, fraction — rather than the zero it is often waved through with; divided by a very low electrical efficiency that makes waste one of the more carbon-intensive rows in Table 2. Waste is also treated as free at the gate, so gate-fee revenue is ignored.

Table 6: Fuel prices and carbon contents. *Published* is `technology-data` at tag v0.13.2; *used here* is what this note assumes. Values in italics differ from the published figure and are therefore assumptions of mine. A dash means the source carries no value for that carrier.

	Price, EUR/MWh_{th}		CO_2, t/MWh_{th}	
Carrier	Published	Used here	Published	Used here
Natural gas	24.6	*35.0*	0.198	0.198
Hard coal	9.6	*12.0*	0.336	0.336
Lignite	3.3	*5.0*	0.407	0.407
Fuel oil	52.9	*55.0*	0.257	0.257
Wood chips	13.6	*25.0*	0.367	*0.000*
Uranium	3.4	3.4	–	0.000
Municipal waste	–	*0.000*	–	*0.150*
Wood pellets	13.6	*38.0*	0.367	*0.000*
Biomethane (upgraded biogas)	62.4	*90.0*	–	*0.000*

Reference capacity factors are assumptions. The "typical duty" column of Table 5, and hence the utilisations in Figure 3.1 and the LCoE column of Table 2, are reference values I have chosen: for the variable technologies, roughly what Danish sites deliver; for the dispatchable ones, roughly the duty each is conventionally built for. They are not model results, and no argument in the note depends on their exact values — which is precisely why Figure 4.1 plots the whole range rather than a point.

The qualitative ratings of Table 1 are judgements. They reproduce the ratings used in the course lectures for the eight technologies those covered, and extend them on the same convention to the rest.

The carbon price of 80 EUR per tonne and the electricity price of 60 EUR/MWh are illustrative, used only for the two carbon figures quoted in Section 3 and for the heat pump's running cost respectively.

D.3 The regional inputs of Section 6

These deserve their own subsection, because they are the least measured numbers in the note and the section that uses them is the one most likely to be quoted out of context.

Every input in Table 4 is an assumption of mine. None is a measurement, none is taken from a published regional cost estimate, and the third significant digit of each is meaningless. They are round numbers chosen to be right in direction and roughly right in magnitude, on the reasoning below. Section 6.1 explains why the note works this way rather than quoting published regional levels: those cannot be harmonised across boundaries and currency years, so Section 6 imports inputs and computes its own levels through Equation (5), at this note's boundary and in this note's currency year. Only the deltas come from outside.

Discount rates are 7.0% for Europe — the note's own assumption, unchanged — 6.0% for the United States and 4.5% for China. The direction is well supported: surveyed required returns for mature renewable projects sit somewhat below European levels in the United States, and state-directed lending at administered rates is widely reported as the largest single source of the Chinese advantage in capital-intensive generation. The IEA's Cost of Capital Observatory is the standing source for such comparisons if you want measured figures. The magnitudes here are mine.

Solar capacity factors are the note's Danish reference (11.5%) against 24.0% for a good American south-western site and 17.0% for a mid-latitude northern Chinese one. This is the best-founded row in the table — solar resource maps are not controversial — and it is also the row doing most of the work.

Investment cost indices are set to 1.15 for the United States and 0.65 for China, with Europe at one. The direction of both is well attested and the reasoning is in Section 6.2: the module is a traded good at something close to a world price, so what differs is installation labour, permitting, grid connection, land and, in the American case, tariffs on imported cells. The sizes are illustrative.

Nuclear investment indices carry the widest uncertainty of anything in this note, and Section 6.4 says so in the text rather than burying it here. Reported costs per kW for recent Chinese serial construction are a small fraction of recent Western first-of-a-kind projects, but the accounting is not comparable and the Western projects themselves span a range far wider than the difference shown. The index of 0.35 is a deliberately conservative reading of that literature and should be treated as an illustration of a mechanism, not an estimate.

Gas and carbon prices are 12 and 40 $\mathrm{EUR/MWh_{th}}$ against the note's European 35, and carbon at 80, 0 and 10 EUR per tonne. The gas figures reflect the persistent discount of American Henry

Hub gas to European TTF since 2022 and China's position as a net importer competing for the same cargoes; the carbon figures reflect the absence of a federal American price and the low level at which the Chinese scheme has traded. Round numbers again.

What the section does *not* assume is worth stating too. Investment costs other than solar and nuclear, all lifetimes, all fixed O&M rates, all efficiencies and all capacity factors other than solar are the note's own European values in every column. That is deliberate — it keeps the number of moving parts small enough to attribute — and it means the non-European columns understate the true variation rather than exaggerating it.

D.4 Reproducing all of it

Every figure, table and quoted number in this note is produced by three scripts in the accompanying repository, run in order:

i. `data/prepare.py` fetches the datasets above at their pinned versions and reshapes them.

ii. `pipeline/run_costs.py` computes annuities, levelised costs over a grid of utilisations and discount rates, the cost decomposition and the learning fit.

iii. `pipeline/build.py` turns those results into the figures, the table fragments and a file of macros holding every number quoted in the prose — which is why the text cannot disagree with the figure beside it.

None of this is required reading, and nothing in the note asks you to run anything. It is here so that any number in it can be checked.

D.5 What is not reproducible, and why

Two honest limitations.

First, the note quotes no primary cost estimate of its own. Every cost comes through a compilation, and compilations make choices — about boundaries, about currency years, about which of several published figures to believe — that are not always documented at the level of the individual number. This is the same criticism Section 4.4 makes of everybody else, and it applies here too.

Second, Section 6 compares regions without measuring any of them. There is a good reason for that — a defensible European–American comparison of published levelised costs needs two sources with harmonised boundaries and currency years, and I have not found a pair I would be willing to put my name to — and the section's response is to import inputs rather than answers, so that every level it prints is computed here at one boundary. That makes the *comparison* sound. It does not make the inputs measured, and they are not: they are the assumptions listed in Appendix D.3. What Section 6 can support is a statement about which channel matters and roughly how much. What it cannot support is a statement about what electricity costs in the United States or China, and it does not make one.